

Basic Information

Experiment No: 03

Date: 11/09/2026

Title: Farm Animals Application using Classes and Objects

CO: CO1

Objective / Aim

To design and develop a Java application using **classes and objects** to represent farm animals such as **cow, pig, and horse**, including details about where they stay, what they eat, and the sounds they make.

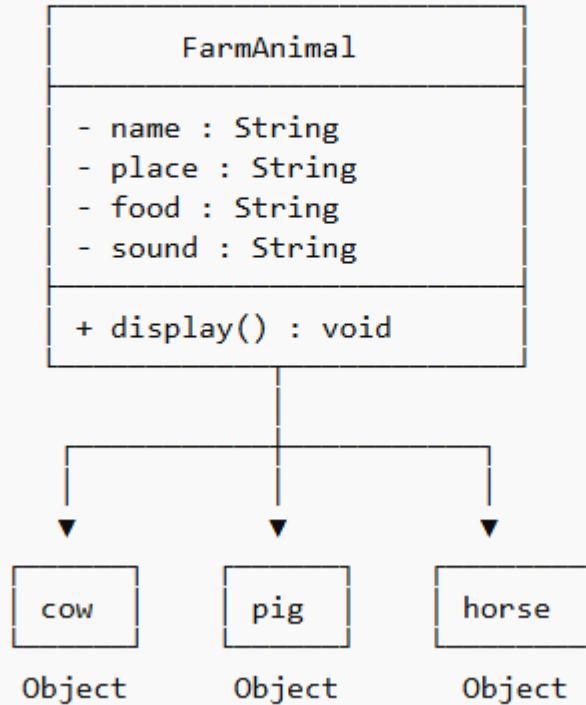
System Requirements

- **Hardware:** Standard PC with Core i3/i5 processor and 4GB/8GB RAM.
- **Software:** Windows/Linux/macOS.
- **Programming Language:** Java.
- **JDK:** JDK 17 or JDK 21.
- **IDE:** Eclipse, IntelliJ IDEA, NetBeans, or VS Code.

Algorithm / Logic Description

1. Create a class named FarmAnimal.
2. Declare the attributes name, place, food, and sound.
3. Create a display() method to display the details of an animal.
4. In the main() method, create an object named cow using the FarmAnimal class.
5. Assign the cow's name, place, food, and sound.
6. Create another object named pig and assign its details.
7. Create another object named horse and assign its details.
8. Call the display() method for each object.
9. Display the details of the cow, pig, and horse on the console.

UML Class Diagram



Source Code:

```
class FarmAnimal {  
    String name;  
    String place;  
    String food;  
    String sound;  
  
    void display() {  
        System.out.println("Animal: " + name);  
        System.out.println("Stays in: " + place);  
        System.out.println("Eats: " + food);  
        System.out.println("Sound: " + sound);  
        System.out.println();  
    }  
}
```

```
}  
}
```

```
public class ClassesAndObjects {  
    public static void main(String[] args) {  
  
        // Creating Cow object  
        FarmAnimal cow = new FarmAnimal();  
        cow.name = "Cow";  
        cow.place = "Cowshed";  
        cow.food = "Grass";  
        cow.sound = "Moo";  
  
        // Creating Pig object  
        FarmAnimal pig = new FarmAnimal();  
        pig.name = "Pig";  
        pig.place = "Pigsty";  
        pig.food = "Grains";  
        pig.sound = "Oink";  
  
        // Creating Horse object  
        FarmAnimal horse = new FarmAnimal();  
        horse.name = "Horse";  
        horse.place = "Stable";  
        horse.food = "Grass";  
        horse.sound = "Neigh";  
  
        // Displaying animal details  
        cow.display();
```

```
        pig.display();  
        horse.display();  
    }  
}
```

Compilation & Execution Commands

Save the file exactly as:

ClassesAndObjects.java

Compile the program:

javac ClassesAndObjects.java

Run the program:

java ClassesAndObjects

Output Screen

```
Animal: Cow  
Stays in: Cowshed  
Eats: Grass  
Sound: Moo
```

```
Animal: Pig  
Stays in: Pigsty  
Eats: Grains  
Sound: Oink
```

```
Animal: Horse  
Stays in: Stable  
Eats: Grass  
Sound: Neigh
```

Result

The Java **Farm Animals application** was successfully implemented using **classes and objects**. The program creates objects for cow, pig, and horse and displays their **name** **place of stay**, **food**, and **sound**.